

EU Semiconductors**EU / Japan semis: Hybrid Bonding – Bullish logic adoption despite HBM uncertainties; Raising Besi PT to €310**

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Bonding technologies are essential to enable advanced packaging, and hybrid bonding (HB) is the ultimate bonding technology, as explained in our reports ([The Bonder War: Part 1 – HBM, Part 2](#) and [Bonder Primer](#)). This report provides our latest view of the current bonding market as well as an update to our TAM forecast.

We see a clear trend for logic to adopt die-to-wafer (D2W) hybrid bonding to increase interconnection speed among logic chiplets and between logic and memory. For logic, we forecast 2028 HB TAM of \$887mn, and 2028 shipment of 237 units (103 prior), to reflect our outlook of stronger adoption at TSMC for SoIC packaging. We expect TSMC's SoIC capacity to grow 8x from 10kwp in 2025 to 85kwp in 2028, for AMD, Apple, Google, and Nvidia applications. Intel is also expected to resume HB capacity expansion from 2027, with the ramp up of Clearwater Forest and Diamond Rapids.

HBM adoption timing is less clear. Samsung may adopt HB in HBM4E (16-hi) in 2027, but if HBM4E turns out to be 12-hi only per [Korean media](#), HB adoption is possibly postponed to 2028 with HBM5, with few (maybe 1) initial layers only while the remaining stays on TCB. For HBM, we forecast 2028 HB TAM of \$489mn, and 2028 shipment of 130 units, based on our view that some portion of HBM4E will adopt hybrid, and HBM5 will migrate to hybrid bonding.

Raising HB TAM forecast. We forecast the hybrid bonder TAM to surge to \$1.4bn in 2028, driven by the adoption in both logic and HBM. Compared to Besi's market forecast, our 2028 unit shipment estimates of 367 units is 24% above the mid case and 8% below the high case. Our HB TAM model is available for [download here](#).

Raising Besi TP to €310, as a technology leader and structural beneficiary of hybrid bonding adoption. Assuming 2028 market share of 76% (80% logic, 70% HBM), Besi's revenue from HB would be \$1.1bn, which is based on unit sales of ~281 units and ASP assumption of \$3.75mn per unit. We raise 2027 / 2028 EPS by 3.6% / 23% respectively and our 2028 EPS is now 16.6% above consensus. Based on 40x (45x prior, cut to reflect weaker visibility on hybrid bonding) 1-year forward P/E and rolling over a quarter, we take Besi's PT up to €310.00 (€280 prior), with 39% potential upside. **Reiterate Outperform rating.**

For DISCO, we expect the real upside would come from hybrid bonding adoption, as dicing / grinding systems with high-cleanness specification would need to be adopted to ensure a particle-free bonding surface, enabling a high yield. Overall setup for DISCO seems favorable into 2027. Current drivers of generative AI in which DISCO partakes is mostly HBM and CoWoS. Whereas from 2027, we expect further adoption of NAND CBA bonding from more players such as Samsung, increased adoption of BSPDN, and die-to-wafer hybrid bonding. In addition, power semis such as SiC is also likely to resume investment. **Reiterate Outperform rating.**

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	20 Jul 2026	Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
			Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
BESI.NA (Besi)	O	EUR	223.50	310.00	57.2%	EUR	1.66	3.83	6.05	134.6	58.4	36.9
OLD				280.00				3.34	5.84			
6146.JP (DISCO)	O	JPY	64,780	85,000	10.9%	JPY	1,246.28	1,733.62	2,127.33	52.0	37.4	30.5
EDME			1,587.83									
JPL			2,545.21									

PRICE TARGET CHANGE / ESTIMATE CHANGE IN BOLD

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate Besi (TP=€310.00) and Disco (TP=¥85,000) Outperform.

DETAILS

Bonding technologies are essential to enable advanced packaging — we have written extensively on the different bonding technologies such as Thermocompression bonding (TCB) as well as hybrid bonding (HB) for both High Bandwidth Memory (HBM) and logic applications, in our previous reports ([The Bonder War: Part 1 — HBM bonding technology shifts](#), [The Bonder War: Part 2 — Path to advanced packaging for logic chips](#) and [The Bonder Primer: Essential enablers of advanced packaging](#)).

This report provides our latest view of the current bonding market as well as an update to our TAM forecast (Exhibit 1). In short,

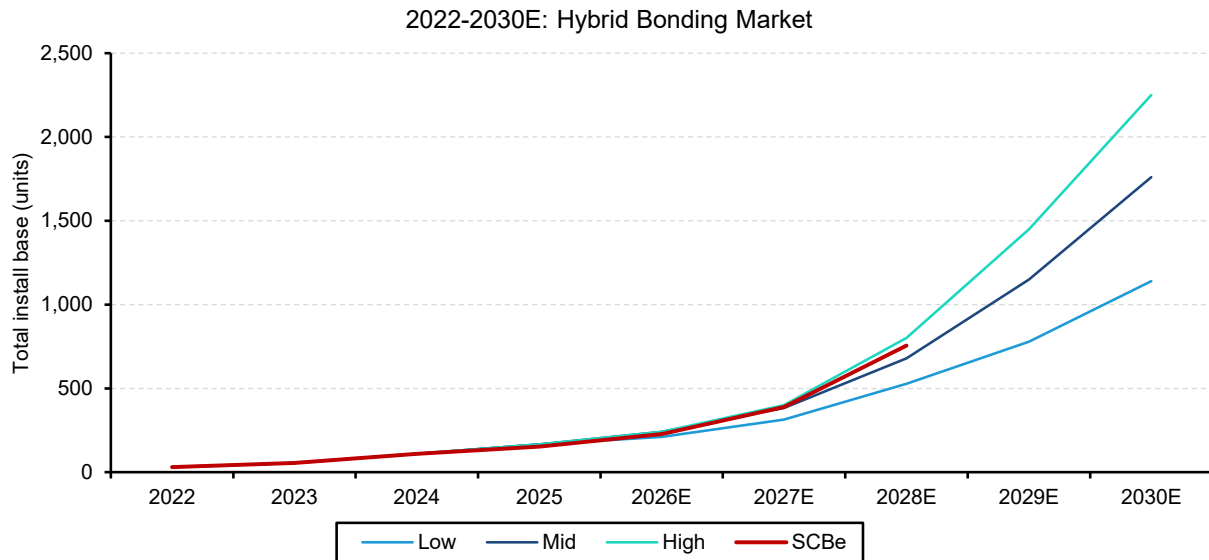
- We see a clear trend for logic to adopt die-to-wafer (D2W) hybrid bonding to increase interconnection speed between logic chiplets and between logic and memory. For logic, we forecast 2028 HB TAM of \$887mn, and 2028 shipment of 237 units (103 prior), to reflect our outlook of stronger adoption at TSMC for SoIC packaging.
- HBM adoption is less clear. While Samsung may adopt HB in HBM4E in 2027, it's possible to be postponed to 2028 with HBM5, with few (maybe 1) initial layer only. We forecast 2028 HB TAM of \$489mn, and 2028 shipment of 130 units, based on our view that some portion of HBM4E will adopt hybrid, and HBM5 will migrate to hybrid bonding.
- We forecast the hybrid bonder TAM to surge to c. \$1.4bn in 2028, driven by the adoption in both logic and HBM. Compared to Besi's hybrid bonding market forecast, we are more optimistic in logic adoption despite being more conservative with HBM adoption of HB, and our 2028 unit shipment estimates of 367 units is 24% above the mid case and 8% below the high case (Exhibit 2).
- For Besi, assuming 2028 market share of 76% (80% logic, 70% HBM), Besi's revenue from HB would be c. \$1.1bn, which is based on unit sales of ~281 units and ASP assumption of \$3.75mn per unit.

The following sections will discuss logic and HBM adoption in detail.

EXHIBIT 1: **Summary of our TCB / HB market forecast.**

Total Market	2024	2025	2026E	2027E	2028E
TC Bonding TAM	654.0	847.2	1,169.7	1,601.4	2,048.9
HBM	399.6	478.1	782.3	1,002.5	1,316.0
Logic	254.4	369.1	387.5	598.9	733.0
% YoY		29.5%	38.1%	36.9%	27.9%
Hybrid Bonding TAM	101.3	99.8	206.3	524.1	1,376.0
HBM	--	--	--	91.8	488.8
Logic	101.3	99.8	206.3	432.4	887.2
% YoY		(1.5%)	106.8%	154.1%	162.5%
Total TCB + HB TAM	755.2	946.9	1,376.0	2,125.5	3,425.0
HBM	399.6	478.1	782.3	1,094.3	1,804.8
Logic	355.6	468.9	593.7	1,031.3	1,620.2
% YoY		25.4%	45.3%	54.5%	61.1%
Besi HB Revenue	101.3	99.8	206.3	488.8	1,052.0
HBM	--	--	--	78.0	342.2
Logic	101.3	99.8	206.3	410.8	709.8
Implied Besi Market Share (HB)	100.0%	100.0%	100.0%	93.2%	76.4%
HBM				85.0%	70.0%
Logic	100.0%	100.0%	100.0%	95.0%	80.0%

Source: Company disclosures, TechInsights, Bernstein estimates and analysis.

EXHIBIT 2: **Our hybrid bonder estimates are between Besi's mid and high case guidance.**

Source: Company disclosures, Bernstein estimates and analysis.

LOGIC: CLEAR TRAJECTORY OF HYBRID ADOPTION

Hybrid bonding is expanding in adoption among the various interconnect technologies, especially for leading-edge chiplet architectures. The key driver remains to be the fact that it enables higher interconnect density while lowering power consumption than with conventional micro-bump based approaches.

We see three primary **use cases** for hybrid bonding:

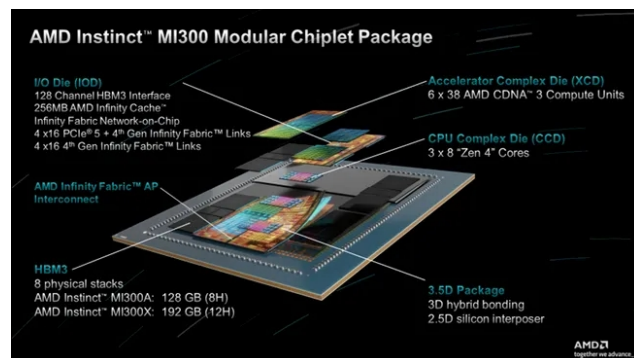
- **XPU-to-XPU chiplet integration (bonding logic on logic):** to improve logic transistor count and boost computing power. Examples of XPU-to-XPU chiplet integrations include AMD's [Instinct GPUs](#) (Exhibit 3) utilizing advanced chiplet architectures, as well as future Nvidia multi-die GPU architectures [such as Feynman GPUs](#). Apple's [M5 Pro](#) uses unique CPU +GPU chiplet integration named Fusion Architecture.
- **XPU+Memory (bonding memory on logic):** to break the memory wall, and there are a few variations:
 - The most prominent is bonding XPU with SRAM, and AMD's 3D V-Cache architecture (Exhibit 4) is a good example of this, and already with a sizeable production volume. EPYC (Milan-X / Genoa-X) and Ryzen X3D (7800X3D / 9800X3D) series have logic die bonded to cache memory die, to realize improved cache capacity as well as lower latency. The Nvidia Groq LPU is also anticipated to use XPU+SRAM integration in the next generation. Intel is also working on their own architecture of HB named Foveros Direct 3D.
 - There is also XPU bonding with DRAM, as shown in China's [LongShine CPU which integrates HBM and DRAM on chip](#). In the LongShine setup, hybrid bonding helps make DRAM behave more like a first-class extension of the processor package rather than a distant off-package resource.
- **CPO (optical communication):** most prominently TSMC's COUPE which integrates optical chip (PIC) and electrical chip (EIC) together with hybrid bonding (Exhibit 5).

For these applications, we expect **hybrid bonding capacity to increase rapidly**, leading to strong growth of hybrid bonding equipment revenue. In terms of capacity, TSMC and Intel are the leaders.

- For TSMC (covered by Mark Li), current use cases are mainly for AMD CPUs and GPUs mentioned above, and some for Apple (M5 Pro / M5 Max). Going forward, we expect TSMC's HB capacity to nearly double for both this year and next year driven by increased adoption, for [Broadcom's](#) 3.5D packaging as well as for Nvidia's Feynman GPUs.

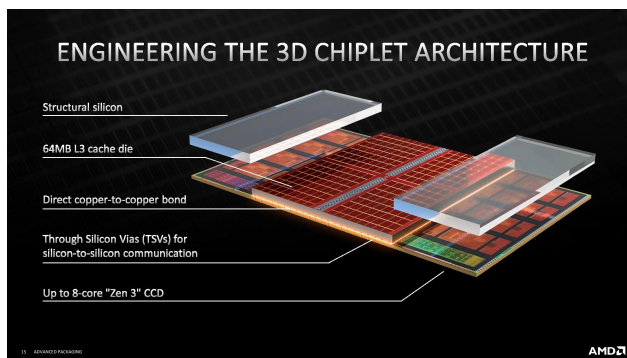
- Intel (covered by Stacy Rasgon) has already deployed hybrid-bonding based architectures through their Foveros Direct 3D platform and remains strategically committed to advanced 3D packaging. Intel's Clearwater Forest is an instance of their HB adoption, and we expect higher adoption going forward as Diamond Rapids and subsequent CPUs start ramping. As for Intel, we don't anticipate much ramp in HB capacity quite yet in 2026 as they already have 30 units of hybrid bonders, and Diamond Rapids seems to be [delayed to 2027](#), but capacity expansion should happen again from 2027 (Exhibit 6-Exhibit 8).

EXHIBIT 3: AMD Instinct GPUs employing XPU-XPU chiplet integration



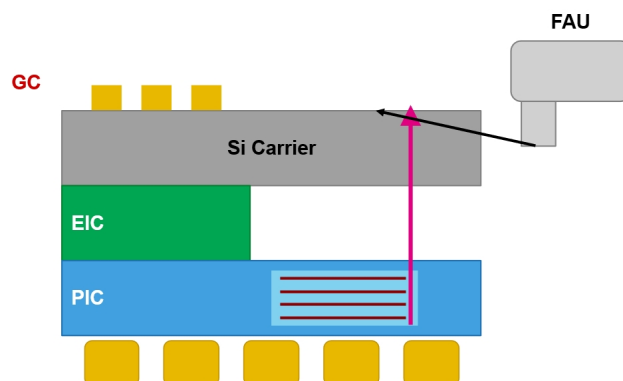
Source: AMD

EXHIBIT 4: AMD 3D V-cache employing XPU-Memory bonding



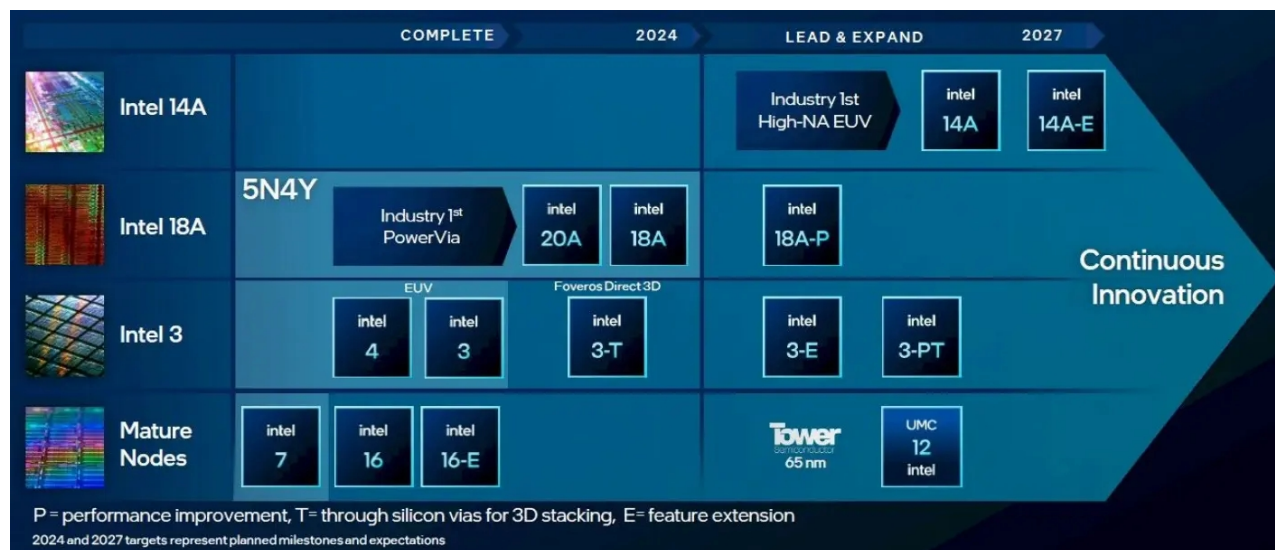
Source: AMD

EXHIBIT 5: Optical interconnects employing hybrid bonding.



Source: Bernstein analysis.

EXHIBIT 6: Intel product roadmap with Foveros Direct 3D



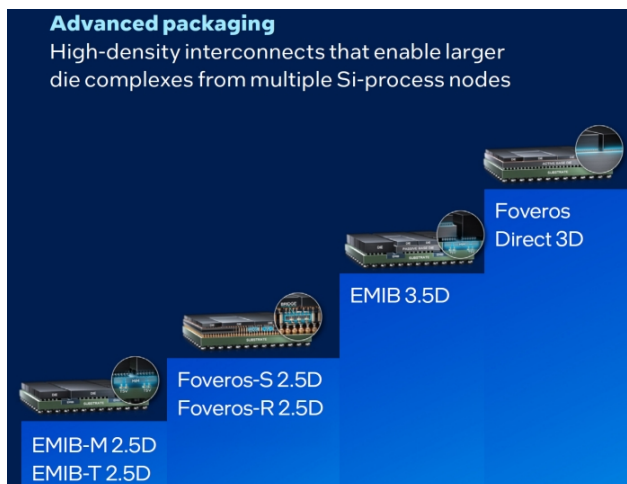
Source: Intel

EXHIBIT 7: Intel Clearwater Forest is Intel's first major CPU that employs hybrid bonding to connect CPU chiplet to base chiplet, before bonding to EMIB substrate.



Source: Intel

EXHIBIT 8: Hybrid bonding (Foveros Direct 3D) is a key technology to enable Intel's product roadmap.



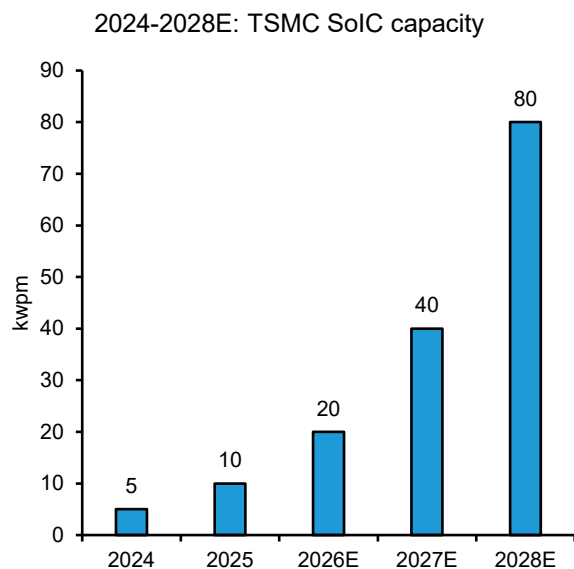
Source: Intel

In terms of TAM, we expect a strong ramp and model the **logic hybrid bonding capacity (installed base) to grow 4.5x to 582 units in 2028 from 137 units in 2025** (Exhibit 10).

- We believe the install base at TSMC could reach ~440 units as AP7 followed by AP8 ramps up — from the current ~65 unit fleet mostly in AP6. AP7 capacity expansion occurring in three phases: 1. Initial SoIC applications 2. HPC chiplet deployment 3. More broad-based adoption such as wafer-level multi-chip module (WMCM) adoption.
- We expect TSMC's capacity grow ~8x from 10kwpm in end 2025 to 80kwpm in 2028 (Exhibit 9), driven by the adoption of AMD, Apple, followed by Google TPU, Groq LPU, and Nvidia GPU (Feynman).
- At a higher level, we estimate that each kwpm SoIC capacity addition requires around 5-6 hybrid bonders, and that the intensity will go down over time as hybrid bonding throughput improves.

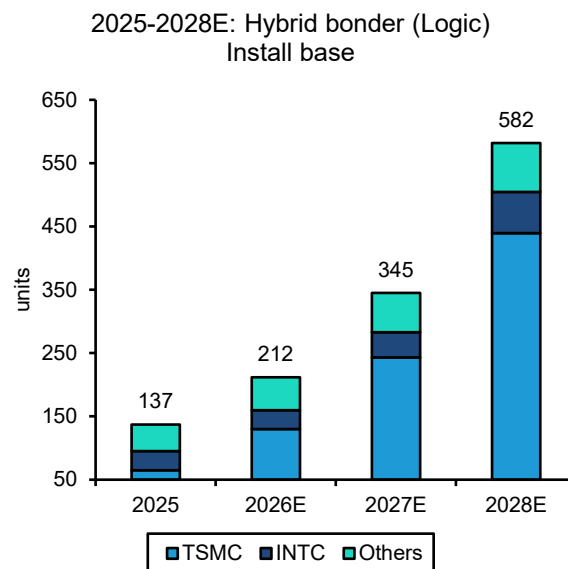
- For Intel, we expect them to resume HB capacity addition from 2027 and model the install base to reach ~65 units in 2028 from the current install base of ~30 units, and others (including Samsung) doubling to ~68 units from ~33 units. As a result, we expect the logic HB TAM to reach \$887mn in 2028, effectively growing eightfold from 2025 (Exhibit 11).

EXHIBIT 9: We expect TSMC's capacity to grow rapidly to 80kwpm in 2028.



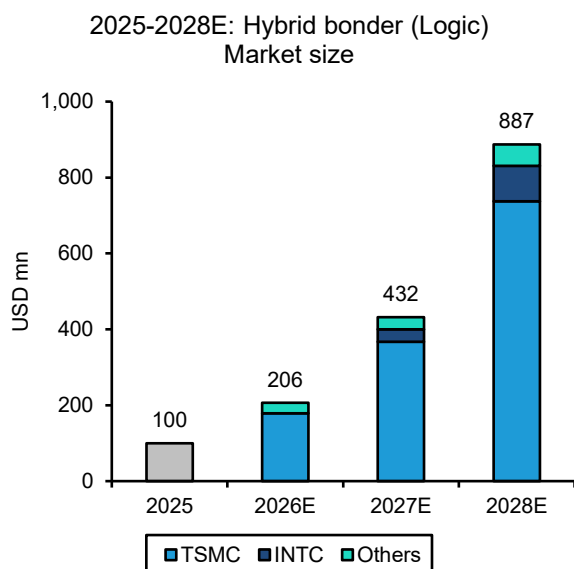
Source: Company disclosures, Bernstein estimates and analysis.

EXHIBIT 10: We expect the install base to grow 4.5x in 3 years.



Source: Company disclosures, Bernstein estimates and analysis.

EXHIBIT 11: As a result, we expect logic hybrid bonder TAM to reach \$887mn in 2028.



Source: Company disclosures, TechInsights, Bernstein estimates and analysis.

HBM: UNCERTAINTIES STILL PRESENT IN HB PENETRATION

While hybrid bonding has become increasingly established in logic applications, adoption within HBM is likely to proceed more gradually. The primary reason is that the current TCB solutions remain highly effective and sufficient for many HBM

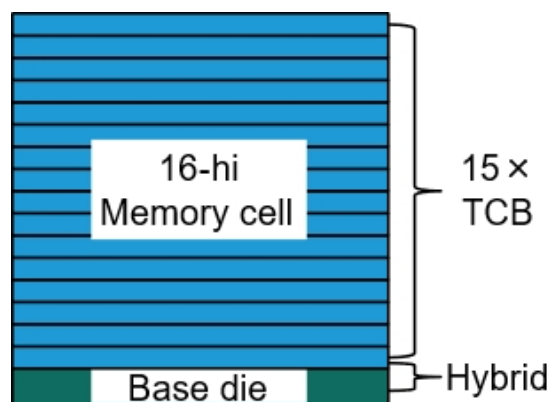
requirements, especially given that 1. JEDEC seems increasingly [likely to relax HBM height standards](#) to ~900µm from the current 775µm, 2. HBM4E seems likely to stay mostly at 12-hi 3. Both [Samsung](#) and [SK Hynix](#) are coming up with new ways to better dissipate heat.

However, in the longer term, we believe factors such as increasing number of layers to stack as well as increase in I/O channels to 4,096 from the current 2,048 eventually would create an incentive to migrate selected layers (Exhibit 12) toward hybrid bonding. We view Samsung to lead the HB adoption followed by SK hynix and Micron, although we believe the majority will continue to be TCB up to 2028. In terms of new capacity addition, we believe TCB shipment should drop after 2028.

Below are our current expectations and potential changes/risks in HBM adoption of HB:

- It looks likely that Samsung could start adoption of HB from 2027 with 16-hi HBM4E, but initially it seems only 1 layer out of 16 layers will use hybrid bonding. However, if HBM 4E only uses 12-hi, then the adoption of HB would be postponed to 2028 with HBM 5.
- Hynix and Micron are likely to follow in 2028, which we expect would be for HBM5. We expect that TCB will still be mainstream technology by then, but TCB shipment will start to decline in 2028 while HB increases steadily.
- As HBM is easier and has lower alignment accuracy requirement, it should be much faster than logic. Every kwpm HBM bonding capacity would only require ~2-3 bonders, less than half of that for logic. Assuming 15% blended penetration in total 1,202kwpm HBM capacity, we derive the HB revenue for HBM bonding to be \$489mn and \$1.8bn in total combined with TCB (Exhibit 14).
- Adding HBM and logic demand, we get total HB TAM of \$1.4bn in 2028 (Exhibit 15).

EXHIBIT 12: **Even in case of HB adoption in 16-hi HBM4E, we believe the adoption would be limited to just 1 layer.**



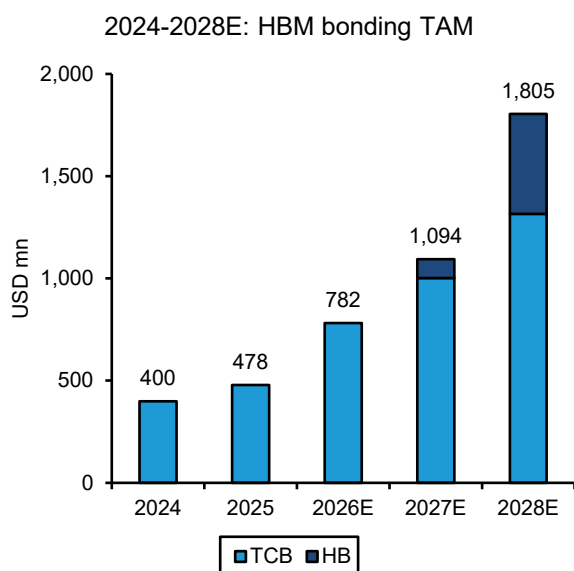
Source: Bernstein analysis.

EXHIBIT 13: **HBM bonding technology roadmap**

Vendor	Scenario 1	Scenario 2
SAMSUNG	HBM4E: 16-hi (HB+TCB) HBM5: 16-hi (HB+TCB)	HBM4E: 12-hi (HB+TCB) HBM5: 16-hi (HB+TCB)
SK hynix	HBM4E: 16-hi (TCB) HBM5: 16-hi (HB+TCB)	HBM4E: 12-hi (TCB) HBM5: 16-hi (HB+TCB)
micron	HBM4E: 16-hi (TCB) HBM5: 16-hi (HB+TCB)	HBM4E: 12-hi (TCB) HBM5: 16-hi (HB+TCB)

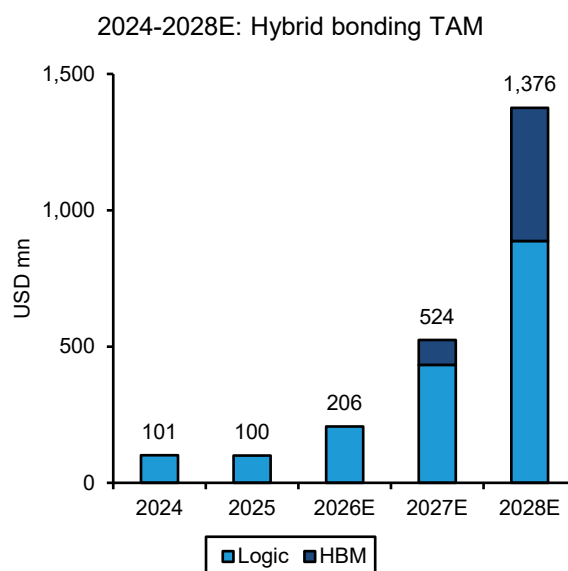
Source: Bernstein estimates and analysis.

EXHIBIT 14: We expect strong HBM bonding growth up to 2028, reaching \$1.8bn.



Source: TechInsights, Bernstein estimates and analysis.

EXHIBIT 15: We expect a strong ramp in hybrid bonding into 2028, reaching \$1.4bn.



Source: Company disclosures, TechInsights, Bernstein estimates and analysis.

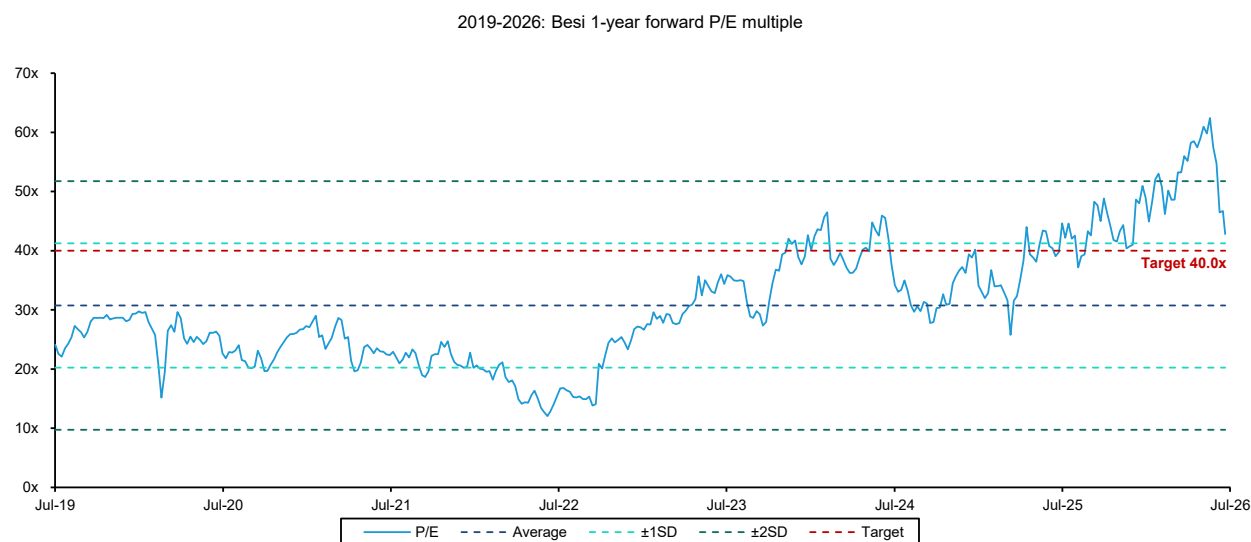
BESI: LIFTING 2028 FORECAST FOR HB ADOPTION AND RAISING OUR PT TO €310.00 (€280 PRIOR)

- For logic, Besi is currently the sole supplier, but we conservatively forecast some share loss and assume 80% for Besi in 2028, as Shibaura, ASMPT, TEL may take some share.
- For HBM, we believe the market share for Besi should be lower in HBM than in logic, as the requirement is lower and the Korean suppliers are likely to participate. SEMES, Hanwha, and Hanmi are all likely to become suppliers. We hence assume only 70% market share in HBM for Besi.
- We forecast Besi's HB revenue at €901mn in 2028 assuming 70% market share in HBM and 80% in logic. On the other hand, we expect the conventional business to recover to €843mn in 2028, from ~€484mn in 2025. In aggregate, we forecast 2028 revenue to reach €1.8bn, which is 11% above consensus. As a result, we expect OP margin to improve further due to scale, reaching 48% in 2028. Our OP estimate for 2028 is now €874mn, 10% above consensus.
- Compared to Besi's long term revenue model, our 2028 numbers of €964mn for "Engine 2 Revenue" (including hybrid bonding and TCB) is 13% above the midpoint, but 23% below for "Engine 1 Revenue" (conventional). Although, we note that this comparison is based on our 2028 numbers, but Besi's long term guidance does not have a target year.
- Based on 40x (45x prior, cut to reflect weaker visibility on hybrid bonding) 1-year forward P/E and rolling over a quarter, we take Besi's PT up to €310.00 (€280 prior), with 39% potential upside. **Reiterate Outperform rating.**

EXHIBIT 16: **Besi: Summary of estimate changes**

Besi: change in estimates	FY25	1Q26	2Q26E	3Q26E	4Q26E	FY26E	FY27E	FY28E
Revenue (EUR mn)								
Bernstein - Old	591.3	166.4	184.9	248.5	199.9	828.4	1,252.6	1,549.9
Growth YoY	-2.7%	8.4%	28.3%	67.8%	50.6%	40.1%	51.2%	23.7%
Bernstein - New	591.3	184.9	266.0	214.0	221.8	886.6	1,237.2	1,806.7
Growth YoY	-2.7%	20.5%	84.5%	44.5%	67.1%	49.9%	39.5%	46.0%
Consensus			251.6	253.1	263.5	942.5	1,306.0	1,622.0
Growth YoY			74.6%	70.9%	98.5%	59.4%	38.6%	24.2%
New vs. old			43.9%	-13.9%	10.9%	7.0%	-1.2%	16.6%
Bernstein vs. consensus			5.7%	-15.4%	-15.8%	-5.9%	-5.3%	11.4%
Operating profit (EUR mn)								
Bernstein - Old	173.1	56.2	63.9	106.2	79.7	322.2	550.1	712.6
Growth YoY	-11.5%	11.1%	62.8%	144.2%	133.8%	86.2%	70.7%	29.5%
Margin	29.3%	33.8%	34.6%	42.7%	39.9%	38.9%	43.9%	46.0%
Bernstein - New	173.1	63.9	114.3	89.5	99.3	367.0	568.1	874.0
Growth YoY	-11.5%	26.4%	191.0%	105.9%	191.1%	112.1%	54.8%	53.9%
Margin	29.3%	34.6%	43.0%	41.8%	44.8%	41.4%	45.9%	48.4%
Consensus			107.5	108.0	113.2	387.5	600.8	795.2
Growth YoY			112.5%	175.1%	160.2%	123.9%	55.0%	32.4%
Margin			42.7%	42.7%	42.9%	41.1%	46.0%	49.0%
New vs. old			78.7%	-15.7%	24.5%	13.9%	3.3%	22.6%
Bernstein vs. consensus			6.3%	-17.1%	-12.3%	-5.3%	-5.4%	9.9%
Basic EPS (EUR)								
Bernstein - Old	1.66	0.54	0.65	1.11	0.83	3.37	5.89	7.75
Growth YoY	-27.7%	-27.4%	63.7%	175.3%	160.8%	102.7%	74.5%	31.7%
Bernstein - New	1.66	0.65	1.20	0.94	1.05	3.86	6.10	9.54
Growth YoY	-27.7%	-12.8%	201.4%	132.4%	228.2%	132.0%	58.0%	56.4%
Consensus			1.13	1.13	1.18	4.00	6.28	8.18
Growth YoY			185.2%	179.1%	270.3%	140.5%	57.0%	30.3%
New vs. old			84.2%	-15.6%	25.9%	14.4%	3.6%	23.0%
Bernstein vs. consensus			5.7%	-16.8%	-11.4%	-3.6%	-2.9%	16.6%

Source: Company disclosures, Bloomberg, Bernstein estimates and analysis.

EXHIBIT 17: **Our target multiple of 40x is set slightly below +1SD of historical range.**

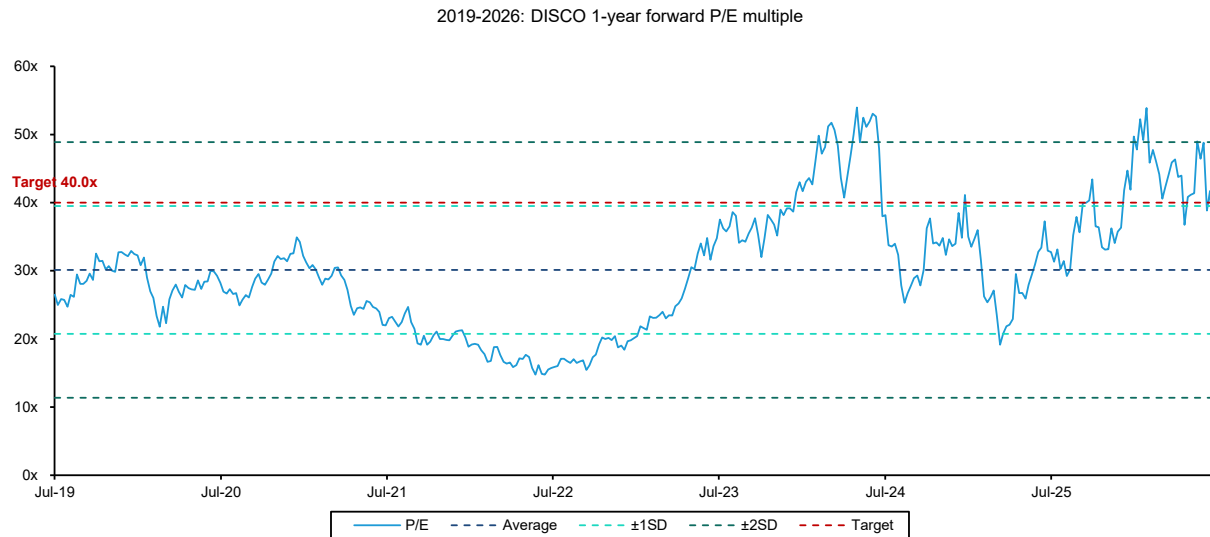
Source: Bloomberg, Bernstein analysis.

DISCO: RIDING THE ADVANCED PACKAGING WAVE

- One other major beneficiary from hybrid bonding migration is DISCO. While broad-based shift towards more advanced packaging has already helped DISCO, we expect the real upside would come from hybrid bonding adoption, as dicing / grinding systems with high-cleanliness specification would need to be adopted to ensure a particle-free bonding surface, enabling a high yield.

- The setup for DISCO seems favorable into 2027. Current drivers of generative AI in which DISCO partakes is mostly HBM and CoWoS. Whereas from 2027, we expect further adoption of NAND CBA bonding from more players such as Samsung, increased adoption of BSPDN, and die-to-wafer hybrid bonding. In addition, power semis such as SiC is also likely to resume investment.
- Disco's valuation is now at 36.5x and has pulled back 32% from the recent peak of 53.9x (Exhibit 18), but we are still positive on the name as we believe their growth drivers are still intact. **Reiterate Outperform.**

EXHIBIT 18: 2019-2026: DISCO 12-month forward P/E



Source: Bloomberg, Bernstein analysis.

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 19: Besi: Income Statement

EUR mn	FY2025	1Q26	2Q26E	3Q26E	4Q26E	FY2026E	FY2027E	FY2028E
Revenue	591.3	184.9	266.0	214.0	221.8	886.6	1,237.2	1,806.7
Gross profit	374.3	117.4	172.6	138.9	146.4	575.3	815.6	1,199.2
SG&A	-120.2	-30.1	-35.0	-29.6	-28.3	-123.0	-148.5	-195.1
R&D	-81.0	-23.4	-23.3	-19.8	-18.9	-85.3	-99.0	-130.1
Other opex	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Operating profit	173.1	63.9	114.3	89.5	99.3	367.0	568.1	874.0
D&A	33.7	9.8	7.6	7.6	7.5	32.5	29.8	29.3
EBITDA	206.8	73.7	121.9	97.1	106.8	399.5	597.9	903.3
Non-OP income / (expense)	-19.1	-5.2	-7.5	-6.0	-6.2	-24.9	-34.8	-50.8
Pretax Income	153.9	58.7	106.8	83.5	93.0	342.1	533.3	823.3
Income tax expense	-22.3	-7.2	-13.0	-10.2	-11.3	-41.7	-65.0	-100.3
Net Income	131.6	51.6	93.8	73.3	81.7	300.4	468.3	722.9
Diluted EPS (EUR)	1.66	0.65	1.20	0.94	1.05	3.83	6.05	9.47
Margins								
Gross profit	63.3%	63.5%	64.9%	64.9%	66.0%	64.9%	65.9%	66.4%
Operating profit	29.3%	34.6%	43.0%	41.8%	44.8%	41.4%	45.9%	48.4%
EBITDA	35.0%	39.9%	45.8%	45.4%	48.2%	45.1%	48.3%	50.0%
Profit before tax	26.0%	31.8%	40.1%	39.0%	41.9%	38.6%	43.1%	45.6%
Net income	22.3%	27.9%	35.3%	34.3%	36.8%	33.9%	37.9%	40.0%
Growth YoY								
Revenue	-2.7%	28.3%	79.6%	61.2%	33.3%	49.9%	39.5%	46.0%
Gross profit	-5.5%	28.0%	84.2%	68.1%	37.8%	53.7%	41.8%	47.0%
Operating profit	-11.5%	62.8%	162.7%	162.6%	76.6%	112.1%	54.8%	53.9%
EBITDA	-7.8%	58.2%	139.2%	125.1%	61.5%	93.2%	49.7%	51.1%
Profit before tax	-18.3%	61.8%	182.5%	188.4%	82.8%	122.2%	55.9%	54.4%
Net income	-27.7%	63.7%	192.6%	190.1%	90.8%	128.2%	55.9%	54.4%

Source: Company disclosures, Bernstein estimates and analysis.

EXHIBIT 20: Besi: Balance Sheet

EUR mn	FY2022	FY2023	FY2024	FY2025	FY2026E	FY2027E	FY2028E
Cash and cash equivalents	491.7	188.5	342.3	373.0	327.2	340.1	429.2
Accounts receivable	170.0	178.1	218.2	173.7	256.6	358.1	522.9
Inventories	92.1	92.5	103.3	104.1	137.3	158.3	188.2
Other current assets	182.9	229.2	334.6	206.3	282.2	282.2	282.2
Total current assets	936.7	688.3	998.4	857.0	1,003.4	1,138.8	1,422.5
Property, plant and equipment, net	114.5	131.2	141.5	158.8	156.2	153.5	151.2
Other non-current assets	84.0	77.1	94.6	97.9	91.9	91.6	91.3
Total assets	1,135.2	896.6	1,234.5	1,113.7	1,251.5	1,383.8	1,665.1
Accounts payable and accrued expenses	105.6	101.3	131.7	56.5	78.4	106.3	153.1
ST Debt and Lease Obligations	5.7	6.9	6.7	0.0	0.0	0.0	0.0
Other current liabilities	32.6	29.0	28.6	97.8	130.8	130.8	130.8
Total current liabilities	143.9	137.2	167.0	154.3	209.3	237.1	283.9
LT Debt and Lease Obligations	322.8	297.4	525.7	507.0	508.1	508.1	508.1
Other non-current liabilities	39.9	40.6	40.6	36.0	34.8	34.8	34.8
Total liabilities	506.7	475.1	733.2	697.4	752.2	780.0	826.8
Shareholders' equity	628.5	421.4	501.3	416.4	499.2	603.8	838.2
Total liabilities & shareholders' equity	1,135.2	896.6	1,234.5	1,113.7	1,251.5	1,383.8	1,665.1

Source: Company disclosures, Bernstein estimates and analysis.

EXHIBIT 21: **Besi: Cash Flow Statement**

EUR mn	FY2022	FY2023	FY2024	FY2025	FY2026E	FY2027E	FY2028E
Net Income	240.6	177.1	182.0	131.6	300.4	468.3	722.9
D&A	23.0	25.7	28.6	33.7	32.5	29.8	29.3
Changes in working capital	-21.6	-26.8	-39.1	-14.4	-58.6	-94.4	-147.6
Other adjustments	29.8	32.6	29.6	27.2	28.3	32.0	49.4
Cash flow from operations	271.9	208.6	201.1	178.1	302.5	435.8	654.0
Capex	-28.4	-28.0	-31.5	-41.8	-27.0	-27.0	-27.0
Other	44.7	-44.9	-105.0	154.8	-80.0	0.0	0.0
Cash flow from investments	16.3	-72.9	-136.5	113.0	-107.0	-27.0	-27.0
Cash flow from financing	-247.7	-439.8	88.7	-261.3	-241.1	-395.8	-537.9
Net effect of exchange rate changes	-0.2	1.0	0.6	0.8	-0.2	0.0	0.0
Net change in cash	40.3	-303.2	153.8	30.7	-45.8	12.9	89.1

Source: Company disclosures, Bernstein estimates and analysis.

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VALUATION METHODOLOGY**BE Semiconductor Industries NV**

For Besi, we set a €310 price target using a P/E multiple of 40x against our forward Q5-Q8 EPS estimate.

Disco Corp

We set a price target of ¥85,000 using a P/E multiple of 40x against our forward Q5-Q8 EPS estimate.

RISKS**BE Semiconductor Industries NV**

Downside risks to our price target for Besi include short-term — lower than expected or delayed adoption of hybrid bonding and prolonged delayed recovery to the mainstream assembly market into late 2025 or even 2026 — as well as longer-term — losing market share in hybrid bonding or lack of take-up of Besi's TCB offering.

Disco Corp

There are a number of downside risks to our DISCO target price: (1) Overcapacity of AI-related production such as GPUs and HBM resulting in both multiple contraction and earning power (2) Changes in competitive environment (3) Unfavorable exchange rate fluctuation with downside risk to its revenue.

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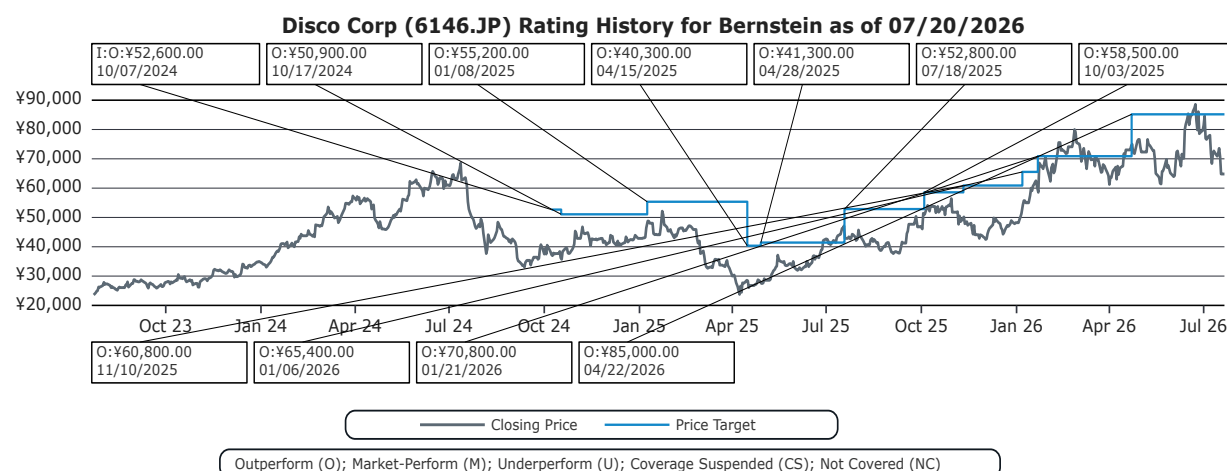
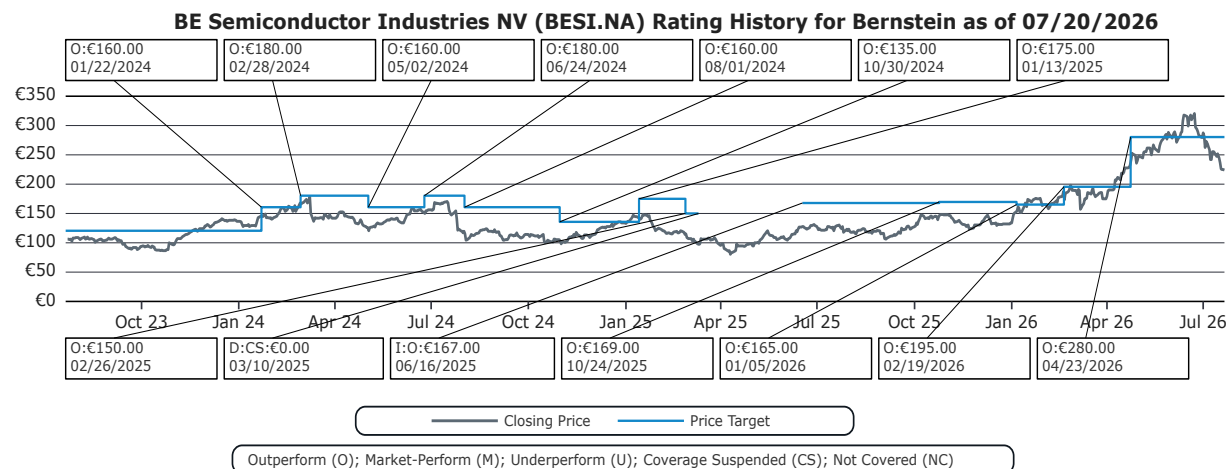
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